

final gaps in localisation can be disproportionately difficult and expensive to fill.

The same challenge appears in semiconductors. The Kae report says India's semiconductor journey is now pivoting from decades of chip design towards end-to-end capability—from IP and fabrication units to OSAT.

It notes that India is initially focusing on areas such as 28-65 nm nodes, power chips and compound semiconductors such as GaN rather than immediately competing at the most advanced logic nodes.

AFTER THE PROTOTYPE

For all the excitement around India's deep-tech ecosystem, the biggest question is whether these companies can cross the gap between invention and industrialisation.

Capital is particularly important. A software start-up can potentially build and iterate relatively cheaply. A space, semiconductor or aerospace start-up may need millions of dollars before it even has a commercially viable product.

Experts say the first breakthrough is not revenue or profitability but whether the technology can become a "founding block" for the future of that sector.

That changes the investment equation.

Founders may need to think in 5-, 10- or even 20-year horizons. Investors have to understand that product validation, certification, manufacturing scale and customer adoption can precede meaningful financial returns by years.

Then there is the customer.

A semiconductor start-up can develop a chip, but an OEM must be willing to qualify it. An aerospace start-up can build a prototype, but certification and production capacity have to follow.



"Pre-2021-2022, nobody was even interested in semiconductors. That changed once the ISM and DLI schemes were announced to encourage start-ups to own IP-designed products"

HAREESH CHANDRASEKAR
CO-FOUNDER AND CEO, AGNIT SEMICONDUCTORS



"Greater localisation across the value chain, driven by strategic priorities, global competitiveness, policy reforms and a maturing talent pool are behind the shift"

P.S. EASWARAN
PARTNER & LEADER OF SUPPLY CHAIN PRACTICE, DELOITTE INDIA

A robotics company can demonstrate an autonomous machine, but manufacturers must be willing to deploy it at scale.

ADOPTION IS A CHALLENGE

That's why the next phase of India's deep-tech story may be less about how many start-ups are being funded and more about how many technologies are actually being deployed.

The Kae report's robotics findings offer a useful warning. India has only about four robots per 10,000 manufacturing workers, compared with roughly 400 in China, while domestic supplier depth remains limited and imported components raise costs. So while the opportunity is enormous, adoption and industrial capability remain bottlenecks.

According to Khandelwal, the technology being developed in India already exists globally, and India is still about 15 years behind in deeptech. Closing that gap will require moving much faster rather than following a linear path of catching up.

That is ultimately what makes India's current deep-tech moment different, and more difficult, than the preceding start-up boom.

The first generation proved India could build businesses at Internet scale. This generation is attempting to prove that India can own the technology, manufacture it and build industries around it.

The journey from Make in India to Invent in India has begun. But invention is only the first milestone. The real test will be whether today's deep-tech start-ups can move from laboratories and prototypes to factories, customers, and markets. **BT**

@PalakAgarwal64